

Database Technologies – Assignment 5

Use appropriate SQL functions to solve following queries. Note: To solve below queries, use “hr” database

1. Display all the first names in lower case and the last names in upper case as the single column separated with ' - '. Provide the meaningful name to the column.

```
SELECT CONCAT(LOWER(FIRST_NAME), ' - ', UPPER(LAST_NAME)) AS FIRST_Name  
FROM employees;
```

```
W2_99546_Pratiksha>SELECT CONCAT(LOWER(FIRST_NAME), ' - ', UPPER(LAST_NAME)) AS FIRST_Name  
-> FROM employees;  
+-----+  
| FIRST_Name |  
+-----+  
| ellen - ABEL |  
| sundar - ANDE |  
| mozhe - ATKINSON |  
| shelli - BAIDA |  
| amit - BANDA |  
| elizabeth - BATES |  
| sarah - BELL |  
| david - BERNSTEIN |  
| laura - BISSOT |  
| harrison - BLOOM |  
| hermann - BROWN |  
| alexis - BULL |  
| alexis - BULL |  
| matthew - WEISS |  
| jennifer - WHALEN |  
| david - WILLIAMS |  
| neena - YANG |  
| eleni - ZLOTKEY |  
+-----+  
107 rows in set (0.00 sec)
```

2. Display the first word of the job title. (Hint : Use substring_index).

```
SELECT SUBSTRING_INDEX(JOB_TITLE, ' ',1)AS first_word_of_job FROM jobs;
```

```
W2_99546_Pratiksha>use hr_db;  
Database changed  
W2_99546_Pratiksha>SELECT SUBSTRING_INDEX(JOB_TITLE, ' ',1)AS first_word_of_job FROM jobs;  
+-----+  
| first_word_of_job |  
+-----+  
| Public |  
| Accounting |  
| Administration |  
| President |  
| Administration |  
| Accountant |  
| Finance |  
| Human |  
| Programmer |  
| Marketing |  
| Marketing |  
| Public |  
| Purchasing |  
| Purchasing |  
| Sales |  
| Sales |  
| Shipping |  
| Stock |  
| Stock |  
+-----+  
19 rows in set (0.01 sec)
```

3. Display the First name, last name and length of first name for the employees whose last name contains character 'b' anywhere after 3rd position.

```
SELECT FIRST_NAME, LAST_NAME, LENGTH(FIRST_NAME)
```

FROM employees

WHERE LAST_NAME LIKE '---%B%';

```
W2_99546_Pratiksha>SELECT FIRST_NAME, LAST_NAME, LENGTH(FIRST_NAME) AS first_name_length
-> FROM employees
-> WHERE POSITION('b' IN LAST_NAME) > 3;
+-----+-----+-----+
| FIRST_NAME | LAST_NAME | first_name_length |
+-----+-----+-----+
| Gerald     | Cambrault | 6                 |
| Nanette    | Cambrault | 7                 |
| Nancy      | Gruenberg | 5                 |
| Susan      | Jacobs    | 5                 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

4. Display the employee id, first name and last name of the employees whose first or last name includes a space in between.

Select employee_id,first_name,last_name from employees where first_name LIKE '% %' OR last_name LIKE '% %';

```
W2_99546_Pratiksha>Select employee_id,first_name,last_name from employees where first_name LIKE '% %' OR last_name LIKE '% %';
+-----+-----+-----+
| employee_id | first_name | last_name |
+-----+-----+-----+
| 112         | Jose Manuel | Urman     |
+-----+-----+-----+
1 row in set (0.00 sec)
```

5. Display first name, salary, and round the salary to thousands.

SELECT FIRST_NAME,SALARY,ROUND(SALARY,-3)FROM EMPLOYEES AS ROUNDSALARY;

```
W2_99546_Pratiksha>SELECT FIRST_NAME,SALARY,ROUND(SALARY,-3)FROM EMPLOYEES AS ROUNDSALARY;
+-----+-----+-----+
| FIRST_NAME | SALARY   | ROUND(SALARY,-3) |
+-----+-----+-----+
| Steven     | 24000.00 | 24000             |
| Neena      | 17000.00 | 17000             |
| Lex        | 17000.00 | 17000             |
| Alexander  | 9000.00  | 9000              |
| Bruce      | 6000.00  | 6000              |
| David      | 4800.00  | 5000              |
| Valli      | 4800.00  | 5000              |
| Diana      | 4200.00  | 4000              |
| Nancy      | 12008.00 | 12000             |
| Daniel     | 9000.00  | 9000              |
| John       | 8200.00  | 8000              |
| Ismael     | 7700.00  | 8000              |
| Jose Manuel | 7800.00  | 8000              |
| Luis       | 6900.00  | 7000              |
| Den        | 11000.00 | 11000             |
| Alexander  | 3100.00  | 3000              |
| Shelli     | 2900.00  | 3000              |
| Sigal      | 2800.00  | 3000              |
| Jennifer   | 4400.00  | 4000              |
| Michael    | 13000.00 | 13000             |
| Pat        | 6000.00  | 6000              |
| Susan      | 6500.00  | 7000              |
| Hermann    | 10000.00 | 10000             |
| Shelley    | 12008.00 | 12000             |
| William    | 8300.00  | 8000              |
+-----+-----+-----+
107 rows in set (0.01 sec)
```

6. Display employee ID and the date on which he ended his previous job in the format as mentioned. (Format: Thursday, October 27th 2011)

Select employee_id,end_date,DATE_FORMAT(end_date, '%w,%m,%d,%y') AS JOB_END_DATE FROM job_history;

7. Display first name and date of first salary of the employees. (Consider 1st day of month as salary day. HINT: LAST_DAY).

```
SELECT EMPLOYEE_ID,first_name,hire_date,date_add(last_day)(HIRE_date),interval +1
day)as first_salary from employees;
```

```
W2_99546_Pratiksha>SELECT EMPLOYEE_ID,first_name,hire_date,date_add(last_day(HIRE_date),interval +1 day)as first_salary from employees;
```

EMPLOYEE_ID	first_name	hire_date	first_salary
100	Steven	2013-06-17	2013-07-01
101	Neena	2015-09-21	2015-10-01
102	Lex	2011-01-13	2011-02-01
103	Alexander	2016-01-03	2016-02-01
104	Bruce	2017-05-21	2017-06-01
105	David	2015-06-25	2015-07-01
106	Valli	2016-02-05	2016-03-01
107	Diana	2017-02-07	2017-03-01
108	Nancy	2012-08-17	2012-09-01
109	Daniel	2012-08-16	2012-09-01
110	John	2015-09-28	2015-10-01
111	Ismael	2015-09-30	2015-10-01
112	Jose Manuel	2016-03-07	2016-04-01
113	Luis	2017-12-07	2018-01-01
114	Den	2012-12-07	2013-01-01
200	Jennifer	2013-09-17	2013-10-01
201	Michael	2014-02-17	2014-03-01
202	Pat	2015-08-17	2015-09-01
203	Susan	2012-06-07	2012-07-01
204	Hermann	2012-06-07	2012-07-01
205	Shelley	2012-06-07	2012-07-01
206	William	2012-06-07	2012-07-01

107 rows in set (0.02 sec)

8. Display first name and experience of the employees in the format of number of years, months and days completed.

```
SELECT FIRST_NAME, HIRE_DATE, CONCAT(TIMESTAMPDIFF(YEAR, HIRE_DATE,
NOW()),'YEARS',TIMESTAMPDIFF(MONTH, HIRE_DATE, NOW())%12,
'MONTH',TIMESTAMPDIFF(DAY,HIRE_DATE,NOW())%31)AS EXPERINCED FROM EMPLOYEES;
```

```
SELECT FIRST_NAME, HIRE_DATE,
CONCAT(
TIMESTAMPDIFF(YEAR,HIRE_DATE,NOW()), ' YEARS ',
TIMESTAMPDIFF(MONTH,HIRE_DATE,NOW()) % 12, ' MONTHS ',
TIMESTAMPDIFF(DAY,HIRE_DATE,NOW()) % 31, ' DAYS'
) AS EXPERIENCE
FROM EMPLOYEES;
```

```
W2_99546_Pratiksha>SELECT FIRST_NAME, HIRE_DATE,
-> CONCAT(
-> TIMESTAMPDIFF(YEAR,HIRE_DATE,NOW()), ' YEARS ',
-> TIMESTAMPDIFF(MONTH,HIRE_DATE,NOW()) % 12, ' MONTHS ',
-> TIMESTAMPDIFF(DAY,HIRE_DATE,NOW()) % 31, ' DAYS'
-> ) AS EXPERIENCE
-> FROM EMPLOYEES;
```

FIRST_NAME	HIRE_DATE	EXPERIENCE
Steven	2013-06-17	13 YEARS 2 MONTHS 29 DAYS
Neena	2015-09-21	10 YEARS 11 MONTHS 9 DAYS
Lex	2011-01-13	15 YEARS 7 MONTHS 16 DAYS
Alexander	2016-01-03	10 YEARS 8 MONTHS 29 DAYS
Bruce	2017-05-21	9 YEARS 3 MONTHS 21 DAYS
David	2015-06-25	11 YEARS 2 MONTHS 4 DAYS
Valli	2016-02-05	10 YEARS 7 MONTHS 27 DAYS
Diana	2017-02-07	9 YEARS 7 MONTHS 0 DAYS
Nancy	2012-08-17	14 YEARS 0 MONTHS 23 DAYS
Daniel	2012-08-16	14 YEARS 0 MONTHS 24 DAYS
John	2015-09-28	10 YEARS 11 MONTHS 2 DAYS

Pat	2015-08-17	11 YEARS 0 MONTHS 13 DAYS
Susan	2012-06-07	14 YEARS 3 MONTHS 1 DAYS
Hermann	2012-06-07	14 YEARS 3 MONTHS 1 DAYS
Shelley	2012-06-07	14 YEARS 3 MONTHS 1 DAYS
William	2012-06-07	14 YEARS 3 MONTHS 1 DAYS

107 rows in set (0.00 sec)

9. Display the employee id, first name and joining date of the employees who joined in the month of august of any year.

```
SELECT EMPLOYEE_ID, FIRST_NAME, HIRE_DATE FROM EMPLOYEES WHERE
MONTH(HIRE_DATE)=8;
```

W2_99546_Pratiksha>SELECT EMPLOYEE_ID, FIRST_NAME, HIRE_DATE FROM EMPLOYEES WHERE MONTH(HIRE_DATE)=8;

EMPLOYEE_ID	FIRST_NAME	HIRE_DATE
108	Nancy	2012-08-17
109	Daniel	2012-08-16
119	Karen	2017-08-10
129	Laura	2015-08-20
134	Michael	2016-08-26
152	Peter	2015-08-20
158	Allan	2014-08-01
189	Jennifer	2015-08-13
202	Pat	2015-08-17

9 rows in set (0.00 sec)

10. Display the details of employees who joined before the year 2015.

```
SELECT EMPLOYEE_ID, HIRE_DATE FROM EMPLOYEES WHERE HIRE_DATE < '2015-01-01';
```

W2_99546_Pratiksha>SELECT EMPLOYEE_ID, HIRE_DATE FROM EMPLOYEES WHERE HIRE_DATE < '2015-01-01';

EMPLOYEE_ID	HIRE_DATE
100	2013-06-17
102	2011-01-13
108	2012-08-17
109	2012-08-16
114	2012-12-07
115	2013-05-18
120	2014-07-18
122	2013-05-01
133	2014-06-14
137	2013-07-14
141	2013-10-17
145	2014-10-01
156	2014-01-30
157	2014-03-04
158	2014-08-01
174	2014-05-11
184	2014-01-27
192	2014-02-04
200	2013-09-17
201	2014-02-17
203	2012-06-07
204	2012-06-07
205	2012-06-07
206	2012-06-07

11. Display the number of days between system date and 1st January 2011.

```
SELECT TIMESTAMPDIFF(DAY, '2011-01-01', '2026-09-09') AS DIFF;
```

W2_99546_Pratiksha>SELECT TIMESTAMPDIFF(DAY, '2011-01-01', '2026-09-09') AS DIFF;

DIFF
5730

1 row in set (0.00 sec)

12. Display number of employees joined after 15th of any month.

```
SELECT COUNT(*)AS EMPLOYEE_COUNT FROM EMPLOYEES WHERE DAY (HIRE_DATE)>15;
```

```
W2_99546_Pratiksha>SELECT COUNT(*)AS EMPLOYEE_COUNT FROM EMPLOYEES WHERE DAY (HIRE_DATE)>15;
+-----+
| EMPLOYEE_COUNT |
+-----+
|                |
+-----+
1 row in set (0.01 sec)
```

13. Display third highest salary of employees.

```
SELECT DISTINCT SALARY FROM EMPLOYEES ORDER BY SALARY DESC LIMIT 1 OFFSET 2;
```

```
W2_99546_Pratiksha>SELECT DISTINCT SALARY FROM EMPLOYEES ORDER BY SALARY DESC LIMIT 1 OFFSET 2;
+-----+
| SALARY |
+-----+
| 14000.00 |
+-----+
1 row in set (0.00 sec)
```

Assignment 5

Note: To solve below queries, use “spj” database

1. Display all the Suppliers, belonging to cities starting with the letter ‘L’.

```
SELECT * FROM S WHERE CITY LIKE 'L%';
```

```
W2_99546_Pratiksha>SELECT * FROM S WHERE CITY LIKE 'L%';
+----+-----+-----+-----+
| S# | Sname | status | city |
+----+-----+-----+-----+
| S1 | Smith | 20 | London |
| S4 | Clark | 20 | London |
+----+-----+-----+-----+
2 rows in set (0.00 sec)
```

2. Display all the Jobs, with the third letter in JNAME as ‘n’.

```
SELECT * FROM J WHERE JNAME LIKE ' __N%';
```

```
W2_99546_Pratiksha>SELECT * FROM K WHERE JNAME LIKE '__N%';
+-----+-----+-----+
| K# | Jname  | CITY  |
+-----+-----+-----+
| K2 | Punch  | Rome  |
| K4 | Console| Athens|
+-----+-----+-----+
2 rows in set (0.00 sec)
```

3. Display all the Supplier names with the initial letter capital.

```
SELECT CONCAT(UPPER(LEFT(SNAME,1)),LOWER(SUBSTRING(SNAME,2)))AS NAMES
FROM S;
```

```
W2_99546_Pratiksha>SELECT CONCAT(UPPER(LEFT(SNAME,1)),LOWER(SUBSTRING(SNAME,2)))AS NAMES FROM S;
+-----+
| NAMES |
+-----+
| Smith |
| Jones |
| Blake |
| Clark |
| Adams |
+-----+
5 rows in set (0.00 sec)
```

4. Display all the Supplier names in upper case.

```
SELECT UPPER(SNAME)AS NAME FROM S;
```

```
W2_99546_Pratiksha>SELECT UPPER(SNAME)AS NAME FROM S;
+-----+
| NAME  |
+-----+
| SMITH |
| JONES |
| BLAKE |
| CLARK |
| ADAMS |
+-----+
5 rows in set (0.00 sec)
```

5. Display all the Supplier names in lower case.

```
SELECT LOWER(SNAME)AS NAME FROM S;
```

```
W2_99546_Pratiksha>SELECT LOWER(SNAME)AS NAME FROM S;
+-----+
| NAME  |
+-----+
| smith |
| jones |
| blake |
| clark |
| adams |
+-----+
5 rows in set (0.00 sec)
```

6. Display the Supplier names and the lengths of the names.

```
SELECT SNAME,LENGTH(SNAME)AS LENGTH_OF_NAME FROM S;
```

```
W2_99546_Pratiksha>SELECT SNAME,LENGTH(SNAME)AS LENGTH_OF_NAME FROM S;
+-----+-----+
| SNAME | LENGTH_OF_NAME |
+-----+-----+
| Smith | 5 |
| Jones | 5 |
| Blake | 5 |
| Clark | 5 |
| Adams | 5 |
+-----+-----+
5 rows in set (0.00 sec)
```

7. Display the current day (e.g. Thursday).

```
SELECT DAYNAME (CURRENT_DATE());
```

```
W2_99546_Pratiksha>SELECT DAYNAME (CURRENT_DATE());
+-----+
| DAYNAME (CURRENT_DATE()) |
+-----+
| Thursday |
+-----+
1 row in set (0.02 sec)
```

8. Display the minimum Status in the Supplier table.

```
SELECT MIN(STATUS)AS MIN_STATUS FROM S;
```

```
W2_99546_Pratiksha>SELECT MIN(STATUS)AS MIN_STATUS FROM S;
+-----+
| MIN_STATUS |
+-----+
| 10 |
+-----+
1 row in set (0.00 sec)
```

9. Display the maximum Weight in the Parts table.

```
SELECT MAX(WEIGHT)AS MAX_WEIGHT FROM P;
```

```
W2_99546_Pratiksha>SELECT MAX(WEIGHT)AS MAX_WEIGHT FROM P;
+-----+
| MAX_WEIGHT |
+-----+
| 19 |
+-----+
1 row in set (0.00 sec)
```

10. Display the average Weight of the Parts.

```
SELECT AVG(WEIGHT)AS MIN_STATUS FROM P;
```

```
W2_99546_Pratiksha>SELECT AVG(WEIGHT)AS MIN_STATUS FROM P;
+-----+
| MIN_STATUS |
+-----+
| 15.1667 |
+-----+
1 row in set (0.00 sec)
```

11. Display the total Quantity sold for part 'P1'.

```
SELECT SUM(QTY)AS TOTAL_QUANTITY FROM SP WHERE `P#`='P1';
```

```
W2_99546_Pratiksha>SELECT SUM(QTY)AS TOTAL_QUANTITY FROM SP WHERE `P#`='P1';
+-----+
| TOTAL_QUANTITY |
+-----+
|          1000 |
+-----+
1 row in set (0.01 sec)
```

12. Display all the Supplier names (with 'la' replaced by 'ro'). HINT: REPLACE.

```
SELECT SNAME,REPLACE(SNAME, 'LA', 'RO')AS NEW_NAME FROM S;
```

```
W2_99546_Pratiksha>SELECT SNAME,REPLACE(SNAME, 'LA', 'RO')AS NEW_NAME FROM S;
+-----+-----+
| SNAME | NEW_NAME |
+-----+-----+
| Smith | Smith    |
| Jones | Jones    |
| Blake | Blake    |
| Clark | Clark    |
| Adams | Adams    |
+-----+-----+
5 rows in set (0.00 sec)
```